

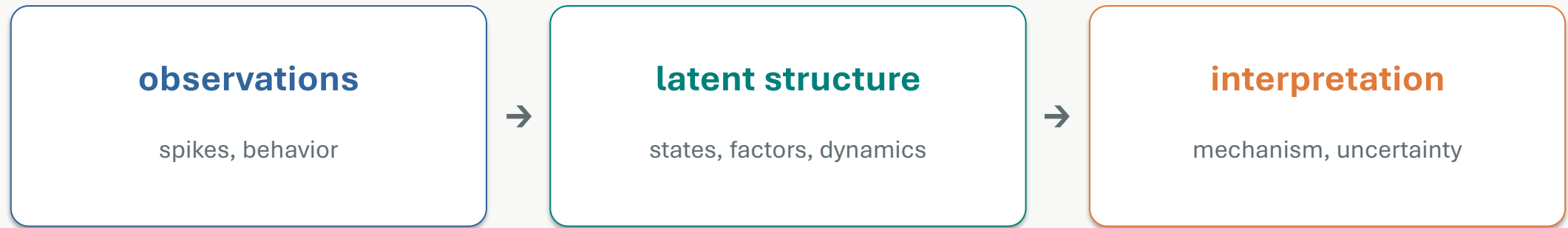
Variational Neural Inference

From probability and gradients to interpretable neural dynamics



One question connects the entire talk

How can noisy, high-dimensional observations
reveal hidden neural computation?



Roadmap: each model adds one assumption

	Probability	uncertainty
	Mixtures	discrete latent classes
	HMMs	temporal dependence
	VAEs	amortized nonlinear inference
	LFADS	recurrent neural dynamics
	gpSLDS	interpretable nonlinear flow

Foundations

Probability, latent variables, gradients, and learning

1

A probabilistic model is a data-generating hypothesis

Observed variables

Measured directly

Examples: spike counts, images, trajectories

Written as x

Latent variables and parameters

Latent z explains hidden structure

Parameters θ specify the model

A joint distribution connects them

Bayes' rule turns a generative model into inference

$$p(z \mid x, \theta) = \frac{p(x \mid z, \theta)p(z \mid \theta)}{p(x \mid \theta)}$$

posterior

what we infer after seeing data

likelihood

how the latent state explains x

prior

belief before observing x

evidence

normalization; often the hard term

Learning and inference are different operations

Inference

Given θ and x



estimate z

Learning

Given x



estimate θ

Prediction

Given new x



predict missing or future data

Gradients tell us how to change parameters

$$\theta_{k+1} = \theta_k - \eta \nabla_{\theta} \mathcal{L}(\theta_k)$$

$\nabla_{\theta} \mathcal{L}$ points toward the steepest local increase in loss

η controls the step size

Backpropagation applies the chain rule through a computation graph

Automatic differentiation computes exact derivatives of the implemented program

Optimization does not remove modeling assumptions

Objective

What quantity should improve?

Parameterization

Which functions can the model express?

Data split

What evidence tests generalization?

Identifiability

Can different parameters explain the same data?

Mixture models and EM

The first nontrivial latent-variable learning problem

2

A mixture model explains heterogeneity

$$p(x_n | \theta) = \sum_{k=1}^K \pi_k \mathcal{N}(x_n | \mu_k, \Sigma_k)$$

π_k

component frequency

μ_k

component center

Σ_k

component shape

Hidden assignment z_n selects the component that generated x_n .

EM alternates posterior inference and parameter learning

E

Expectation step

Compute responsibilities
 $r_{nk} = p(z_n = k \mid x_n, \theta)$



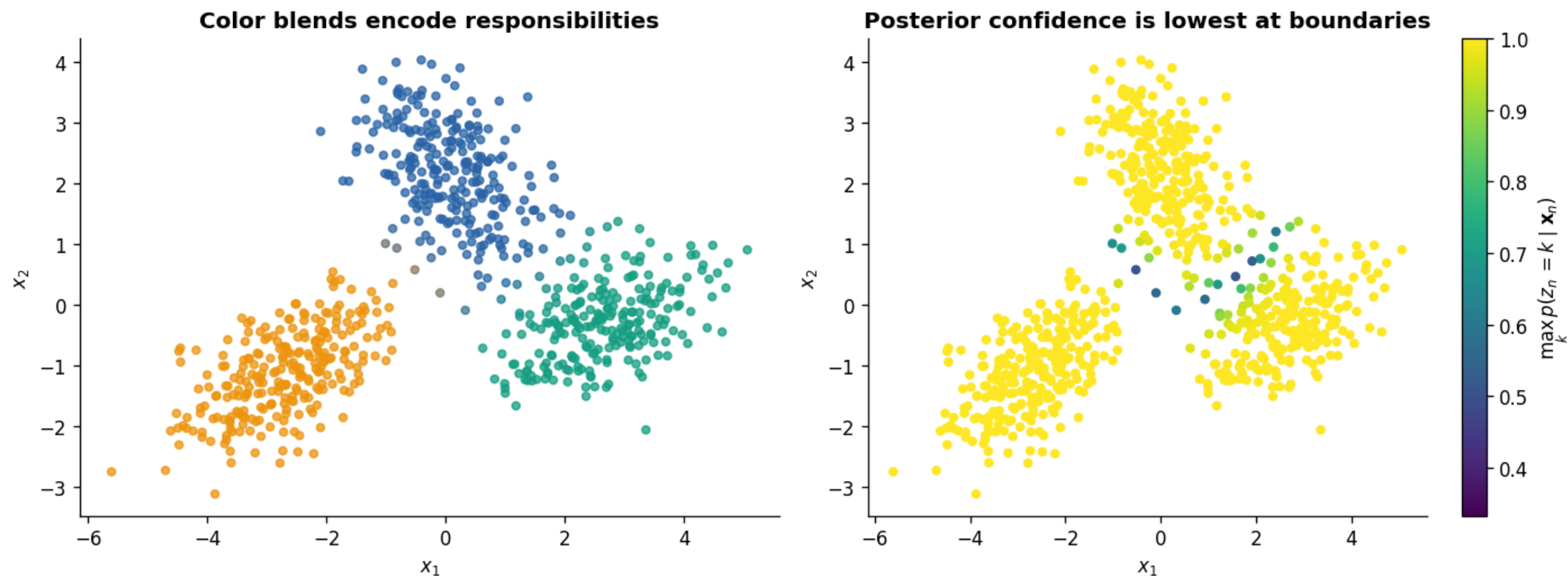
M

Maximization step

Refit π, μ, Σ using
posterior-weighted statistics

Each full iteration does not decrease the data log likelihood.

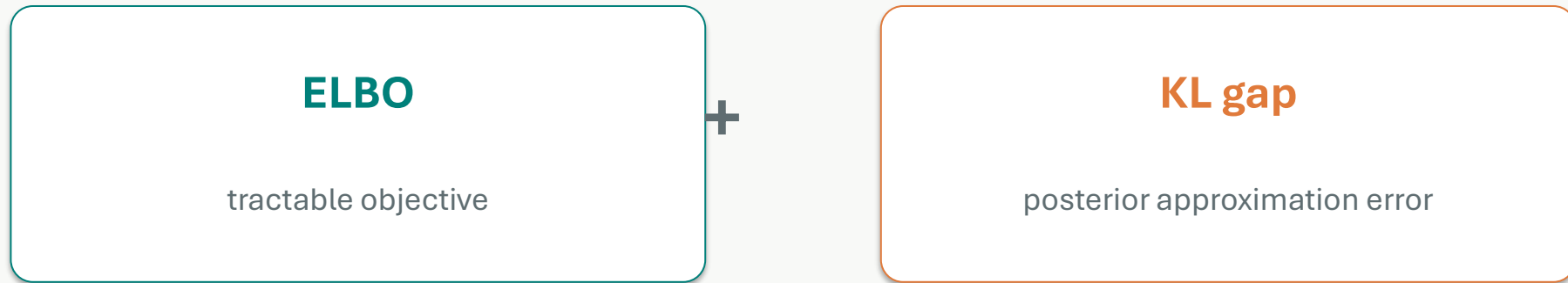
Soft assignments preserve uncertainty



Color mixtures represent posterior responsibilities; the right panel shows assignment confidence.

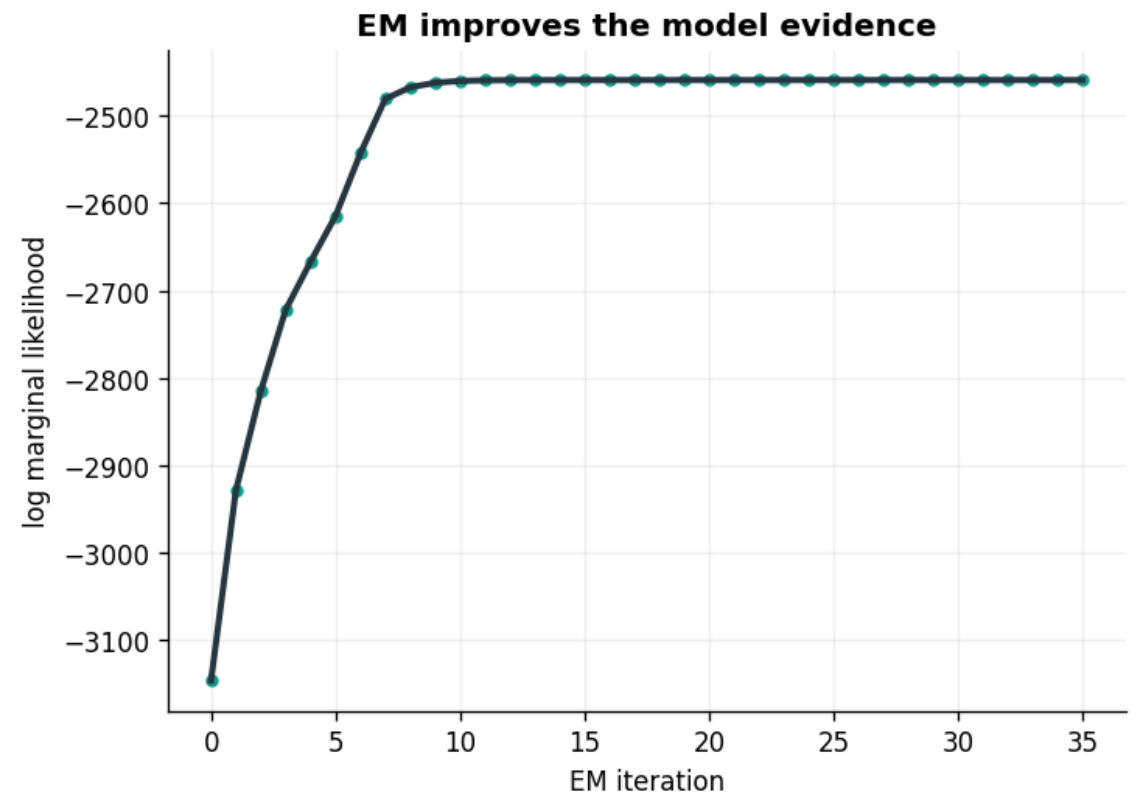
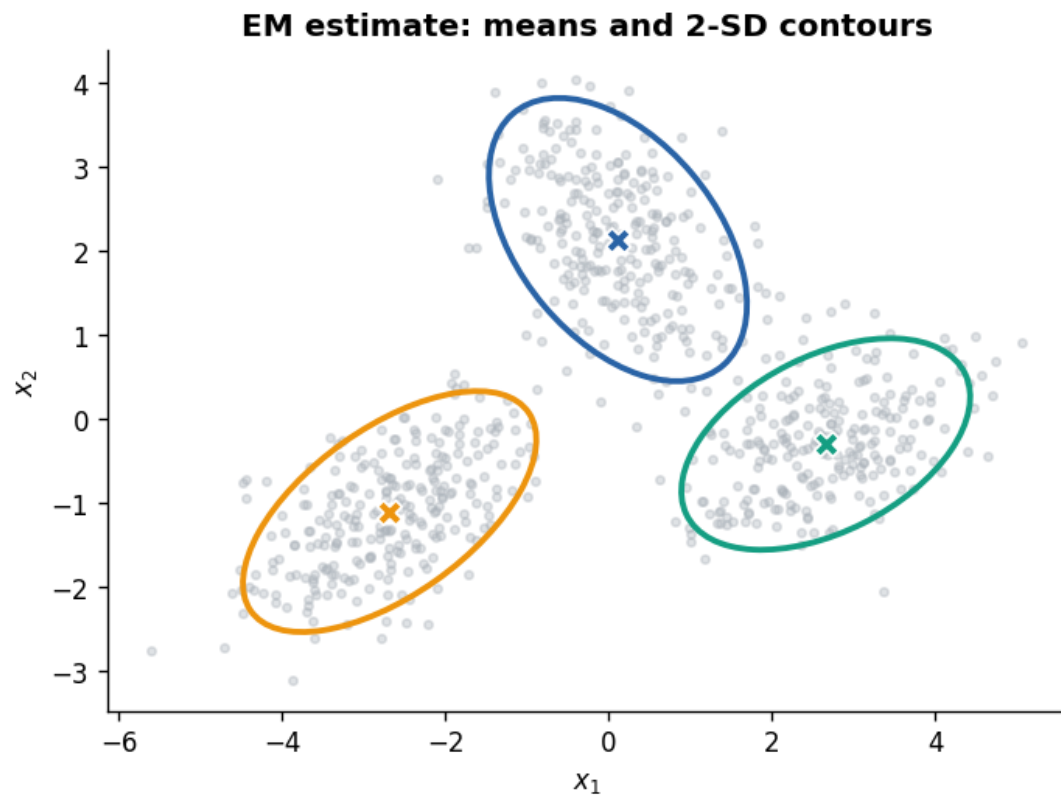
EM is coordinate ascent on a lower bound

$$\log p(x) = \mathcal{L}(q, \theta) + \text{KL}[q(z) \parallel p(z \mid x, \theta)]$$



Exact E-step: $q(z) = p(z \mid x, \theta)$, so the KL gap becomes zero.

EM produces a fitted density, not only labels



Left: recovered Gaussian geometry. Right: monotonic observed-data log likelihood.

What a mixture model cannot represent

Exchangeability

The model gives the same probability after any permutation of observations.



same histogram



different sequence

To model persistence, switching, and prediction, z must depend on its history.

Hidden Markov models

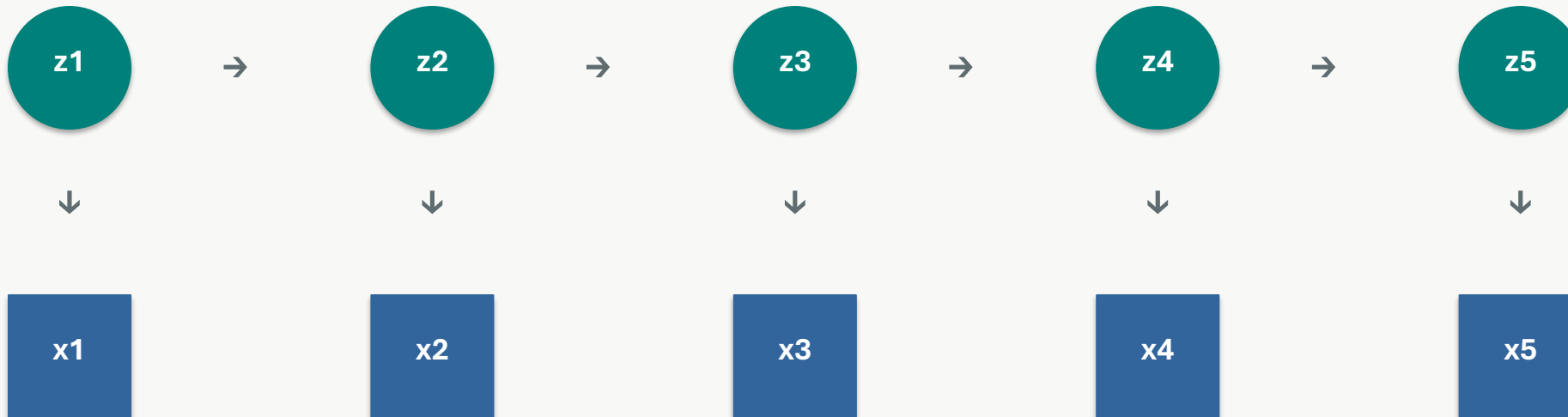
Discrete latent states with temporal persistence

3

An HMM adds a Markov chain to a mixture

$$p(z_{1:T}, x_{1:T}) = p(z_1) \prod_{t=2}^T p(z_t | z_{t-1}) \prod_{t=1}^T p(x_t | z_t)$$

transition model



emission model

Dynamic programming makes exact sequence inference possible

Forward message

$$\alpha_t(k) \propto p(x_t | k) \sum_j \alpha_{t-1}(j) A_{jk}$$

past evidence

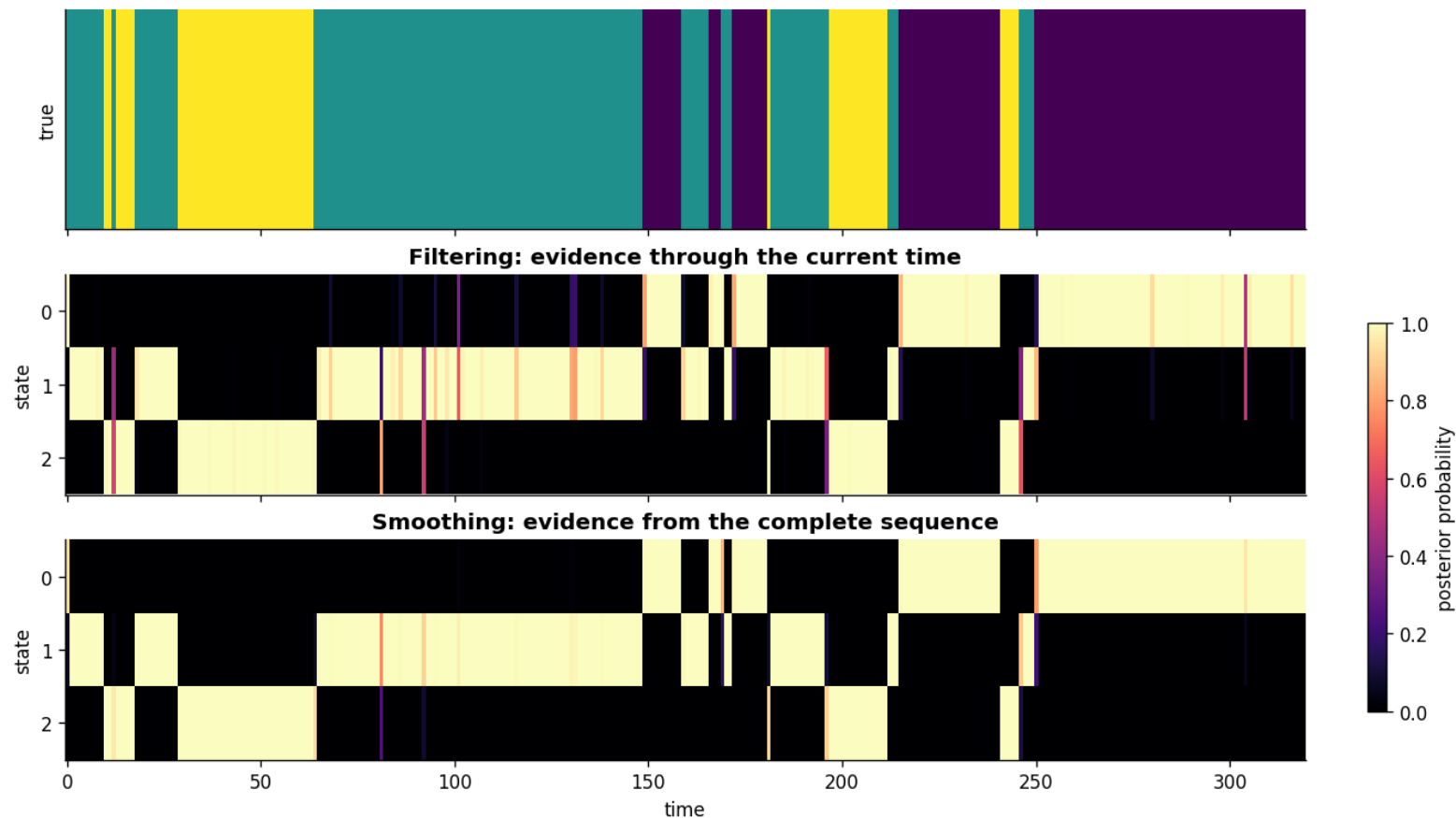
Backward message

$$\beta_t(k) = \sum_j A_{kj} p(x_{t+1} | j) \beta_{t+1}(j)$$

future evidence

smoothed posterior $\propto$ forward $\times$ backward

Filtering and smoothing answer different questions



Filtering uses the past and present; smoothing revises state beliefs using the complete sequence.

Three posterior summaries serve different goals

Marginals

$p(z_t | x_{1:T})$

Uncertainty at each time

Expected occupancy and transitions

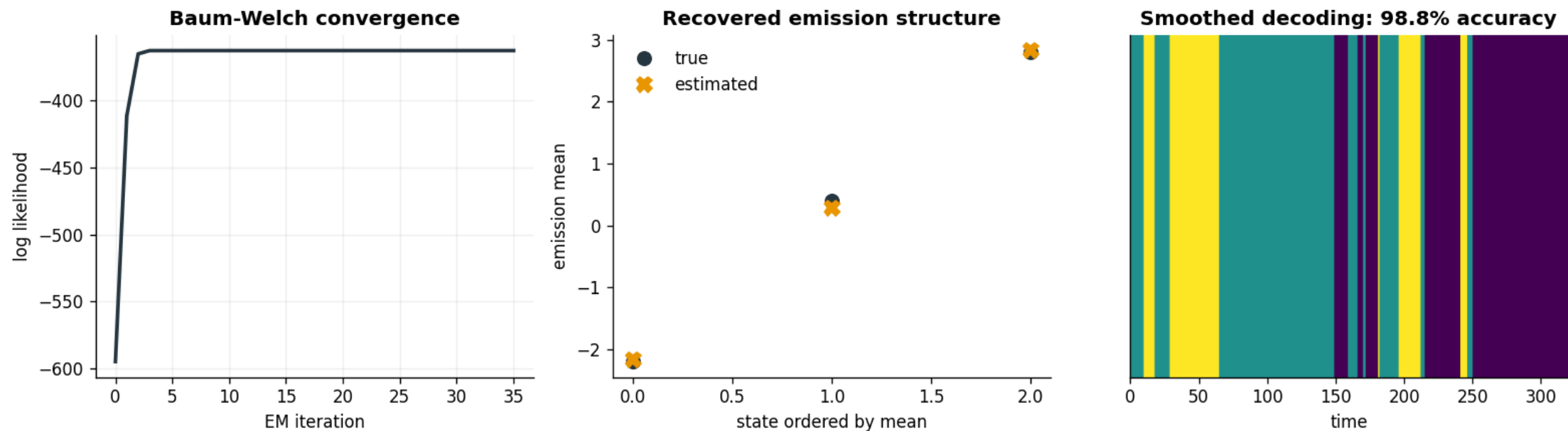
Paths

Viterbi: one MAP sequence

Posterior sampling: distribution over sequences

A single path hides ambiguity

Baum-Welch is EM for an HMM



Forward-backward supplies expected state occupancies and transition counts for the M-step.

HMM assumptions determine the scientific interpretation

Model	Observation / dynamics	Interpretation
Gaussian HMM	state-specific mean/covariance	recurring static regimes
AR-HMM	state-specific autoregression	switching local dynamics
SLDS	continuous latent state + switches	piecewise linear dynamics

Variational inference and VAEs

Approximate posteriors for nonlinear latent-variable models

4

Variational inference turns inference into optimization

$$\log p(x) = \mathcal{L}(q, \theta) + \text{KL}[q(z) \parallel p(z \mid x, \theta)]$$

Choose $q\phi(z \mid x)$

a tractable posterior family

Maximize the ELBO

fit ϕ and model parameters θ

Check the gap

approximation quality is not guaranteed

Amortization learns a reusable inference rule

Classical variational inference

new local parameters for every x_n

optimize separately

accurate but expensive

Amortized inference

shared encoder $q\phi(z | x)$

one forward pass

fast but may introduce amortization gap

The VAE objective balances reconstruction and regularization

$$\mathcal{L} = \mathbb{E}_{q_{\phi}(z | x)}[\log p_{\theta}(x | z)] - \text{KL}[q_{\phi}(z | x) \| p(z)]$$

Reconstruction

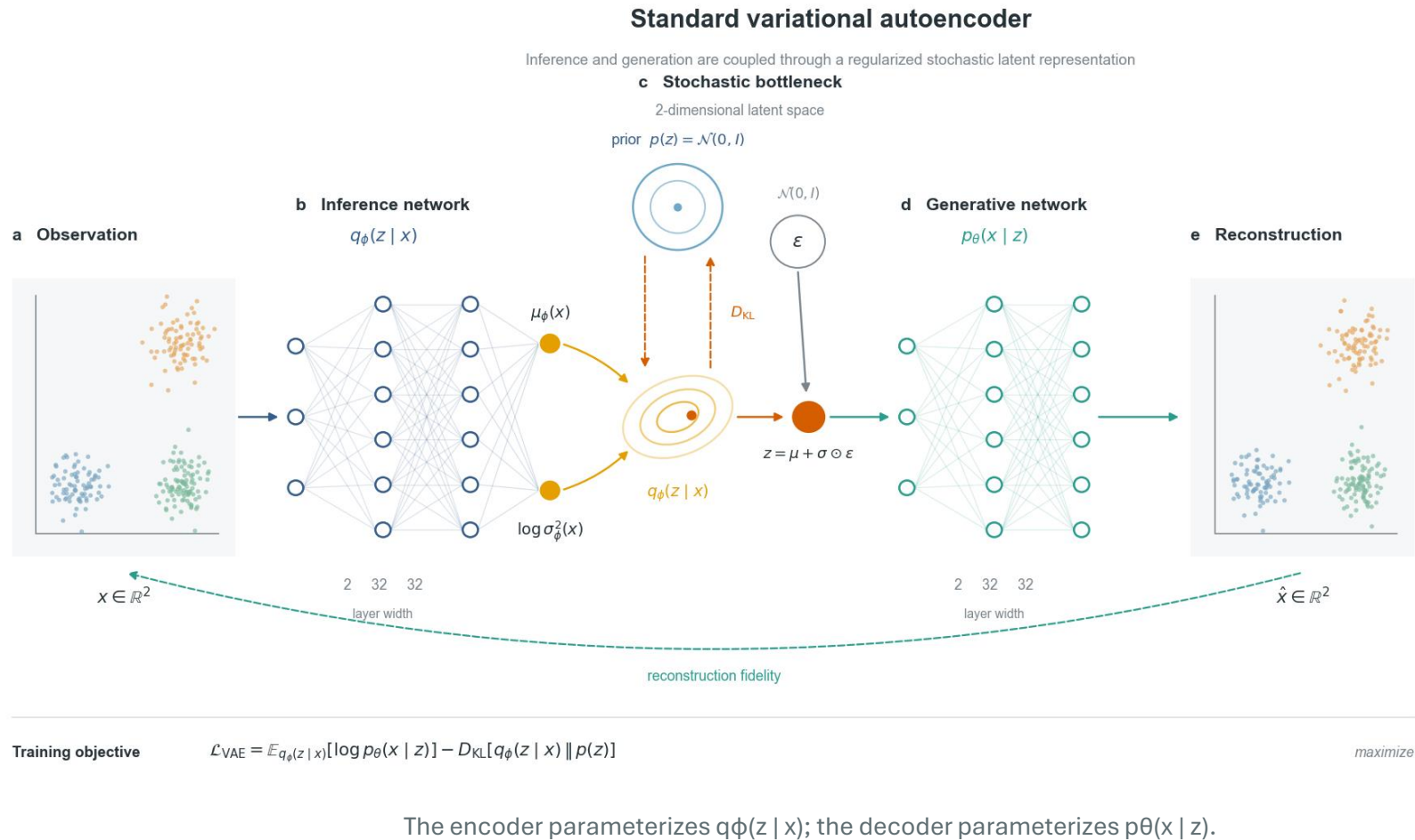
Does z explain x through the decoder?

KL regularization

Does the inferred code remain close to the prior?

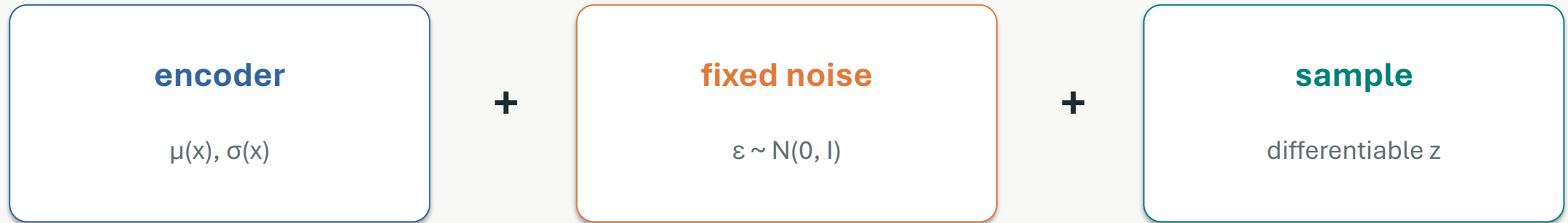
The balance controls information capacity and can cause posterior collapse.

A VAE combines a generative model and an inference network



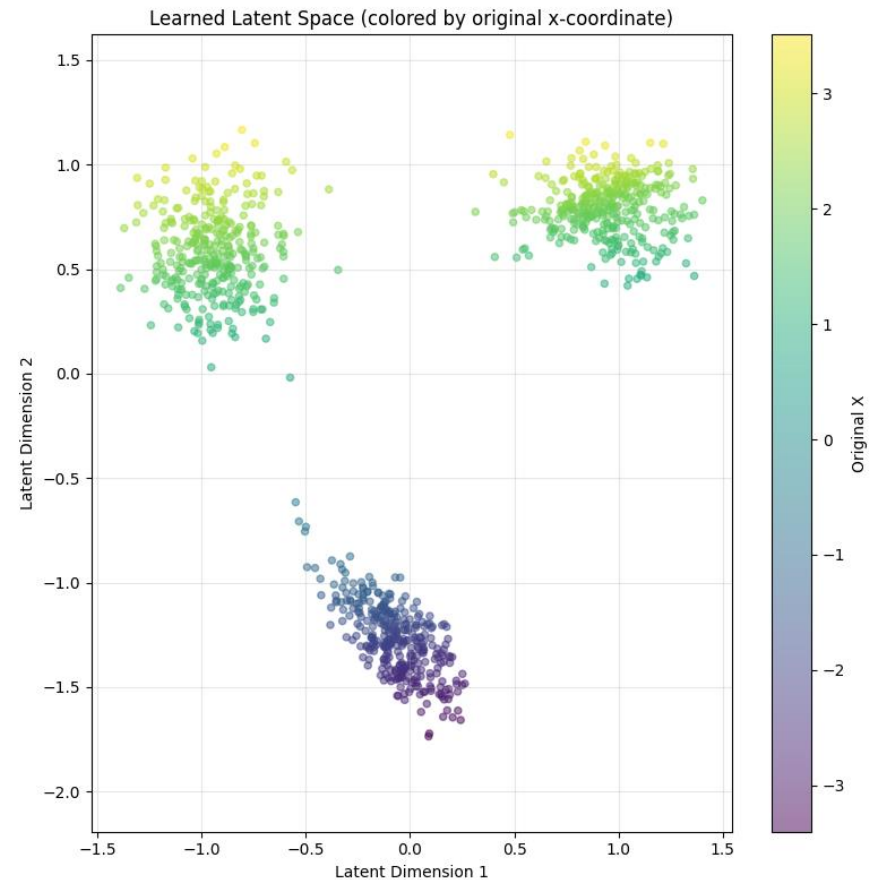
The reparameterization trick differentiates through sampling

$$z = \mu_{\phi}(x) + \sigma_{\phi}(x) \odot \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I)$$



Randomness is moved into ε , leaving a differentiable deterministic transformation.

A latent space is useful only when validated



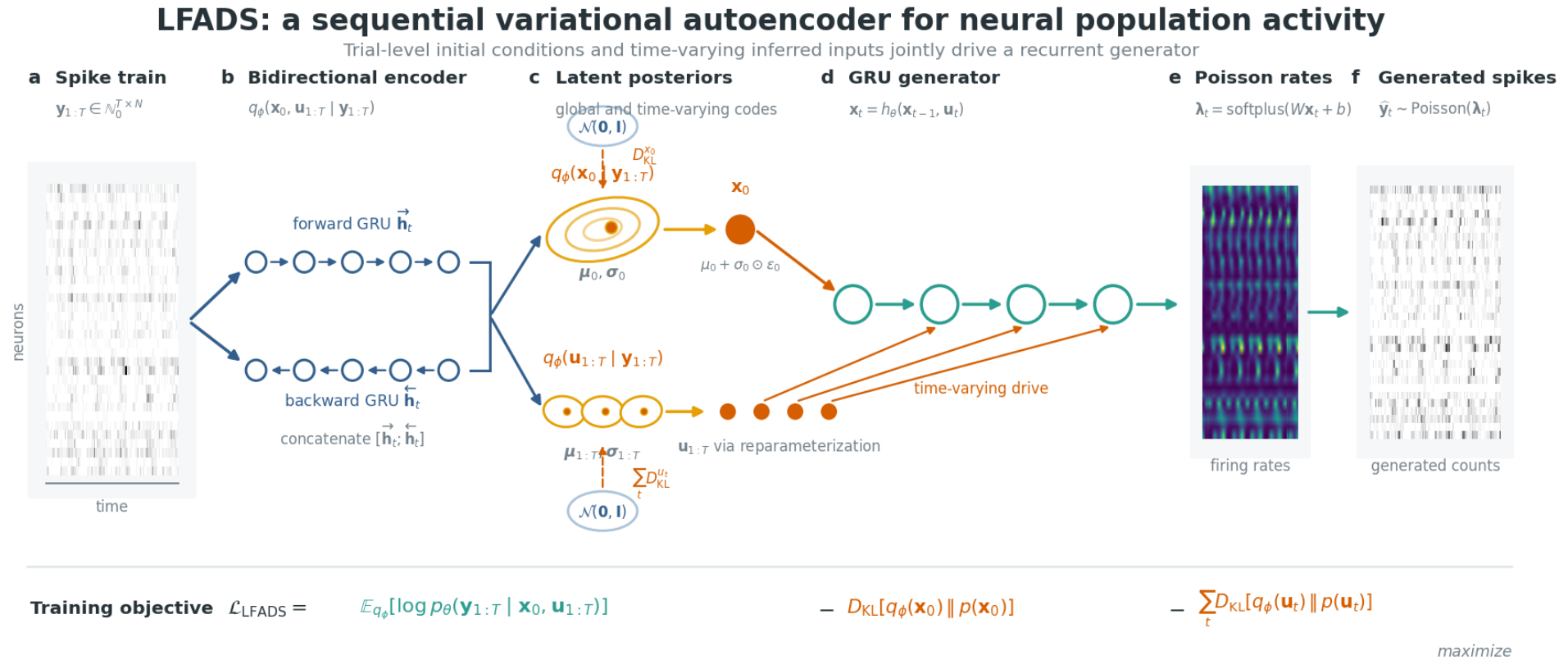
A two-dimensional code can separate structure, but orientation and scale are generally not identifiable.

LFADS

A sequential VAE for latent neural dynamics

5

LFADS separates inferred initial condition, dynamics, and rates



An encoder infers a distribution over initial generator state; recurrent dynamics produce factors and firing rates.

The sequential ELBO assigns uncertainty to a trajectory

Poisson reconstruction

Do inferred rates explain held-out spikes?

Initial-state KL

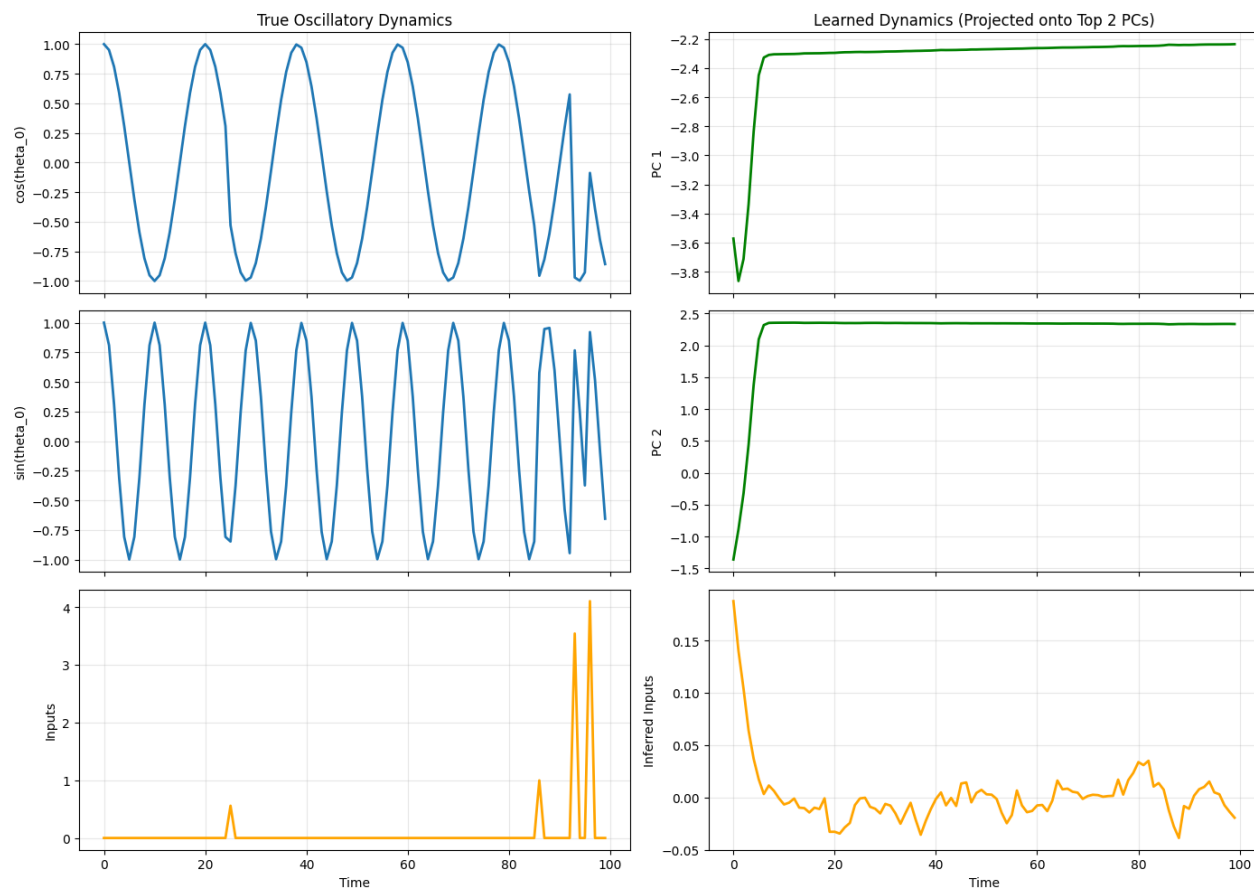
How much trial-specific information enters g_0 ?

Input KL

Are inferred perturbations necessary and plausible?

Scientific danger: inferred inputs can absorb model mismatch.

Learned dynamics must generalize beyond training trajectories



Compare learned and true vector fields, not only reconstructed observations.

gpSLDS

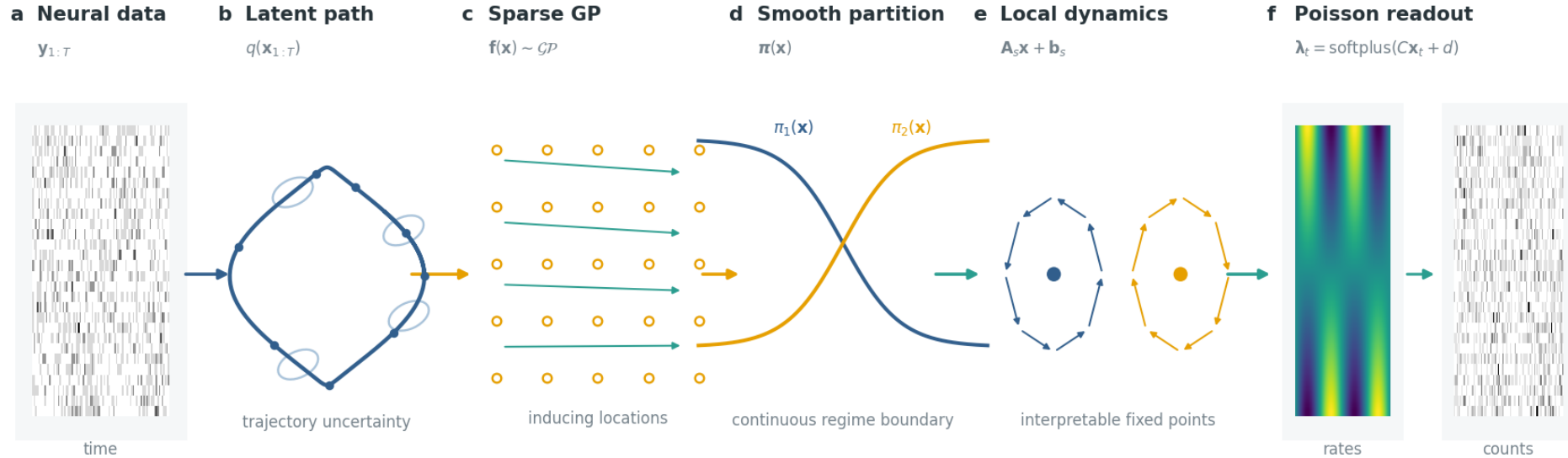
Nonlinear dynamics assembled from interpretable local regimes

6

gpSLDS: nonlinear dynamics with local regimes

Gaussian process switching linear dynamical system

Smooth state-dependent regimes define an uncertain nonlinear latent vector field

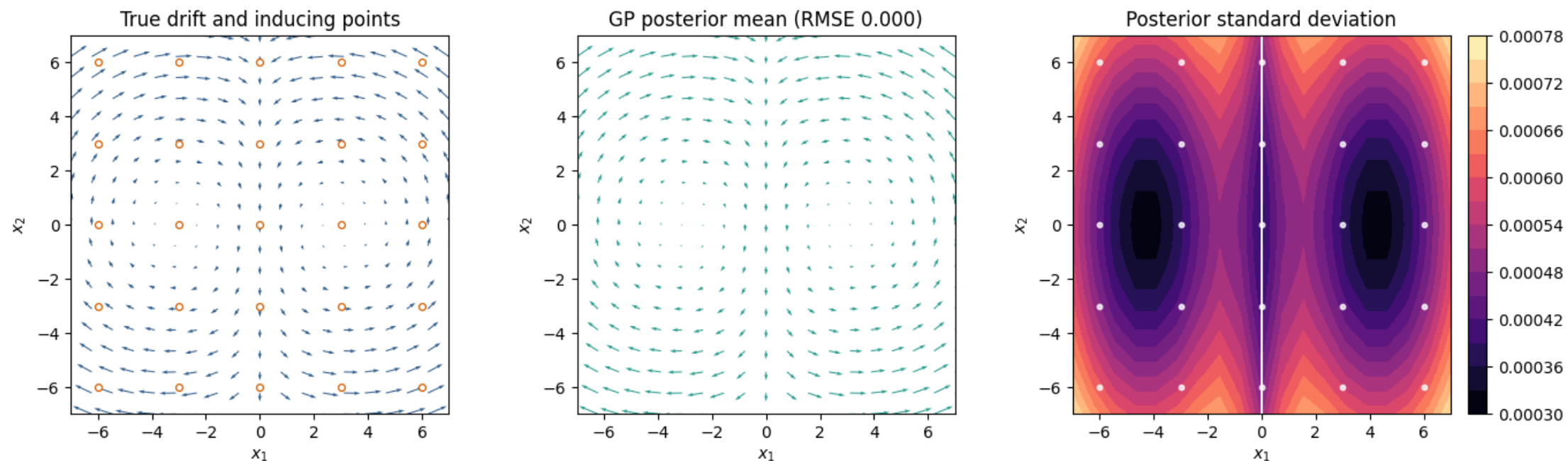


SSL kernel
$$k_{\text{SSL}}(\mathbf{x}, \mathbf{x}') = \sum_{s=1}^S \pi_s(\mathbf{x}) \pi_s(\mathbf{x}') k_s^{\text{lin}}(\mathbf{x}, \mathbf{x}')$$

smooth switching + local linear structure + uncertainty

A smoothly switching kernel blends local linear dynamics within a global nonlinear field.

Local linear structure makes nonlinear dynamics inspectable



Regime weights partition latent space smoothly; local Jacobians support fixed-point and stability analysis.

Choose the simplest model that answers the scientific question

Model	Latent structure	Inference	Best use
GMM	discrete class	exact EM	heterogeneous populations
HMM	discrete sequence	forward-backward	recurring temporal regimes
VAE	continuous code	amortized VI	nonlinear representation
LFADS	recurrent trajectory	sequential VI	population spike dynamics
gpSLDS	continuous + local regimes	variational GP	interpretable nonlinear flow

A practical validation ladder

1

Implementation

synthetic recovery, invariants, numerical stability

2

Predictive

held-out likelihood, spikes, trajectories

3

Robustness

seeds, hyperparameters, latent dimension

4

Scientific

external variables, perturbations, falsifiable predictions

Take-home message

- 1 Inference is the computation that connects a generative hypothesis to data.
- 2 EM, forward-backward, and VAEs share the same posterior-learning logic.
- 3 Temporal structure changes both the algorithm and the scientific meaning of a latent state.
- 4 **Interpretability comes from constrained assumptions plus validation, not from low dimension alone.**

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